

MLFF ADAPT-VERIFY1 adaptive final verification specification

Status: implemented in `mdstats 0.20.127a0`.

Purpose

ADAPT-VERIFY1 closes the adaptive campaign’s deployment-selection loop. It consumes only the fully evaluated admissible candidates already purchased by ADAPT-EVAL1, verifies them in authoritative full-score order, and publishes exactly one target-head model: the first candidate that passes the bounded deployment/NVE gates.

It performs no new target/replay accuracy evaluation and does not restore EVAL-MF screening.

Inputs and ordering

The authoritative parent is `AdaptiveFullEvaluationRecord`. Only its `admissible_candidates` enter verification. Their order is the deterministic EVAL1 full-score order:

1. lower weighted full target/replay force score;
2. lower target force RMSE;
3. lower replay force RMSE;
4. lower original finalist rank;
5. stable checkpoint SHA-256.

By default `[verification].fallback_to_next_full_evaluation_candidate = true`.

The best candidate is verified first. If it fails a hard verification condition, the next already fully evaluated admissible candidate is verified. Verification stops immediately at the first pass. If fallback is disabled, failure of the best candidate is terminal.

Verification matrix and hard gates

Each attempted candidate receives the complete configured verification matrix:

- every `[verification].structures` entry;
- every `[verification].temperatures_kelvin` value;
- the configured timestep, NVE step count, sampling interval, and velocity seed.

A candidate passes only if every case satisfies:

- finite model/trajectory observables;
- absolute energy drift $\leq$ `maximum_energy_drift_ev_per_atom_per_ps`;

- minimum pair distance $\geq$ `minimum_pair_distance_angstrom`;
- maximum force $\leq$ `maximum_force_ev_per_angstrom`.

A failure is a hard rejection of that deployment candidate. Verification does not modify its EVAL1 accuracy score or retroactively change RANK1/STOP1 evidence.

Precision boundary

`[campaign].precision_profile` controls only learned-model arithmetic:

`single` -> FP32 MACE inference and FP32 deployed learned-model bytes
`double` -> FP64 MACE inference and FP64 deployed learned-model bytes

There is no mixed or refine mode. No FP32 candidate is promoted to FP64 during verification or export and no FP64 candidate is demoted.

mdstats-owned scientific arithmetic remains hard-coded FP64 in both modes, including persistent MD state, energy/observable accumulation, drift regression, and reporting/statistical reductions. The verification record exposes both `model_inference_dtype` and `scientific_analysis_dtype`.

Target-head materialization and publication

An EVAL1 winner may be a cross-validation fold run or a final-development run. Historical committee export requires final-development runs and therefore cannot be used to manufacture adaptive selection semantics.

For each candidate, mdstats:

1. authenticates the frozen checkpoint or its qualified storage capsule;
2. reconstructs the exact checkpoint model without dtype-template casting;
3. extracts the target head into an internal verification-only artifact using the selected learned- model dtype;
4. runs bounded NVE on those exact target-head bytes;
5. deletes failed candidate model materializations; and
6. atomically promotes the exact passing bytes into `models/`.

The final deployed model is therefore byte-identical to the artifact that passed NVE. There is no post-verification re-export or dtype conversion.

Adaptive production publishes a single `AdaptiveDeploymentModelRecord`. It does not fabricate a legacy committee around a fold winner. Historical committee evidence remains readable for historical evaluator paths.

Evidence and restart semantics

Each NVE case has a content-addressed identity containing at least:

- adaptive full-evaluation candidate identity;

- checkpoint SHA-256 and target-head model SHA-256;
- verification structure SHA-256;
- temperature, timestep, NVE steps, sampling interval, and velocity seed;
- device and learned-model inference dtype;
- invariant FP64 scientific-analysis dtype;
- acceleration/critical-precision policy identities; and
- verification runtime identity.

Authenticated completed cases are reused after interruption.

AdaptiveVerificationCandidateRecord freezes one candidate’s case digests, model bytes, full score, full target/replay RMSE, dtype boundary, pass/fail outcome, and hard rejection reasons.

AdaptiveVerificationRecord freezes the score-ordered attempted candidate sequence. A successful record must end at exactly one first passing candidate; evidence after a passing candidate is invalid.

AdaptiveDeploymentModelRecord binds the published model bytes to the passing full-evaluation candidate and source checkpoint.

AdaptiveProtocolFreezeRecord binds the production qualification, campaign, full target and true- replay domains, EVAL1 record, VERIFY1 record, deployment model, selected checkpoint, model dtype, and invariant FP64 analysis dtype.

Required final report content

`results/production-verification.json` publishes at least:

- selected EVAL1 finalist rank/batch and run/checkpoint provenance;
- common full target force RMSE;
- common true-replay force RMSE;
- target/replay weights, thresholds, and weighted full score;
- foundation replay RMSE and absolute/fractional degradation diagnostics;
- retained target energy, focus-force, stress, and worst-condition metrics;
- full target/replay domain identities and counts;
- every attempted candidate and its verification case results/rejection reasons;
- verification thresholds and fallback count;
- acceleration/backend identity;
- learned-model inference dtype and FP64 scientific-analysis dtype; and
- exact published model path and SHA-256.

Acceptance criteria

ADAPT-VERIFY1 is qualified only if:

1. candidates are tried strictly in authoritative EVAL1 full-score order;

2. fallback occurs only after a hard verification failure and only among already fully evaluated admissible candidates;
3. no target/replay reevaluation is introduced by fallback;
4. completed NVE cases are restart-reused under exact identities;
5. failed candidate artifacts are not published;
6. the first passing candidate's exact verified target-head bytes are atomically published;
7. `single|double` model dtype is preserved through verification and deployment;
8. mdstats-owned state/analysis remains FP64;
9. a fold-run candidate may be deployed without synthetic committee/final-run substitution; and
10. historical committee/refine/EVAL-MF evidence cannot silently determine an adaptive winner.